Privacy in distributed quantum sensing

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Outlook

I. Statement of the problem

II. When do sensors learn *“too much”*?

III. Privacy in continuous-variable network

In collaboration with:



Jonatan Bohr Brask



Benjamin Larsen



Ulrik Lund Andersen

Technical university of denmark



Masahiro Takeoka

Keio university



Anton L. Andersen



Sean William Moore



Damian Markham

Sorbonne université

Based on:

arXiv:2509.12338

I. Statement of the problem

How can we **ensure** that the **information** processed in a **quantum network** is **secure** from malicious parties?

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Quantum cryptographic ranging

Vittorio Giovannetti[1], Seth Lloyd[1,2], and Lorenzo Maccone[1]
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[1] *Research Laboratory of Electronics*
[2] *Department of Mechanical Engineering*
Cambridge, MA 02139, USA.

Private network parameter estimation with quantum sensors

Nathan Shettell
Centre for Quantum Technologies, National University of Singapore, Singapore 117543, Singapore

Majid Hassani and Damian Markham
LIP6, CNRS, Sorbonne Université, 4 place Jussieu, F-75005 Paris, France

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Now at Leonardo Innovation Labs – Quantum Technologies, Via Tiburtina km 12,400, 00131 Rome, Italy

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Privacy in Networks of Quantum Sensors

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Private and Robust States for Distributed Quantum Sensing

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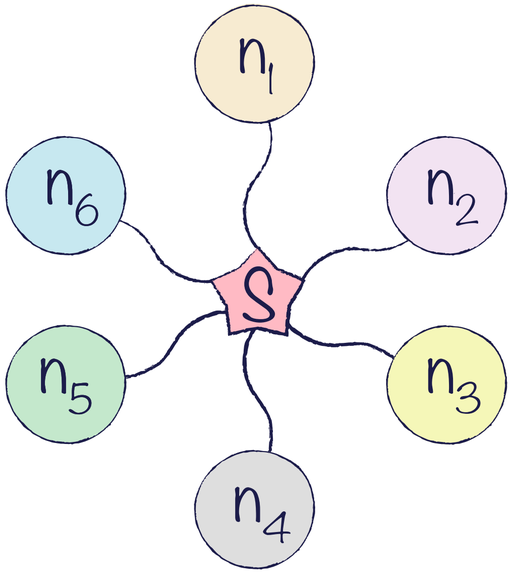
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and many more...

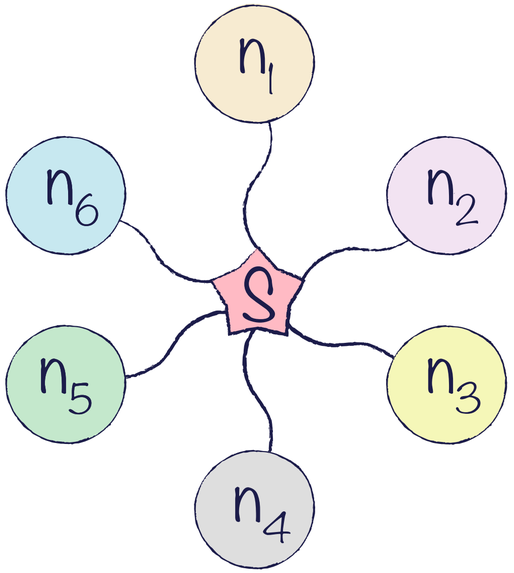
simple example

- Network of quantum sensors:



simple example

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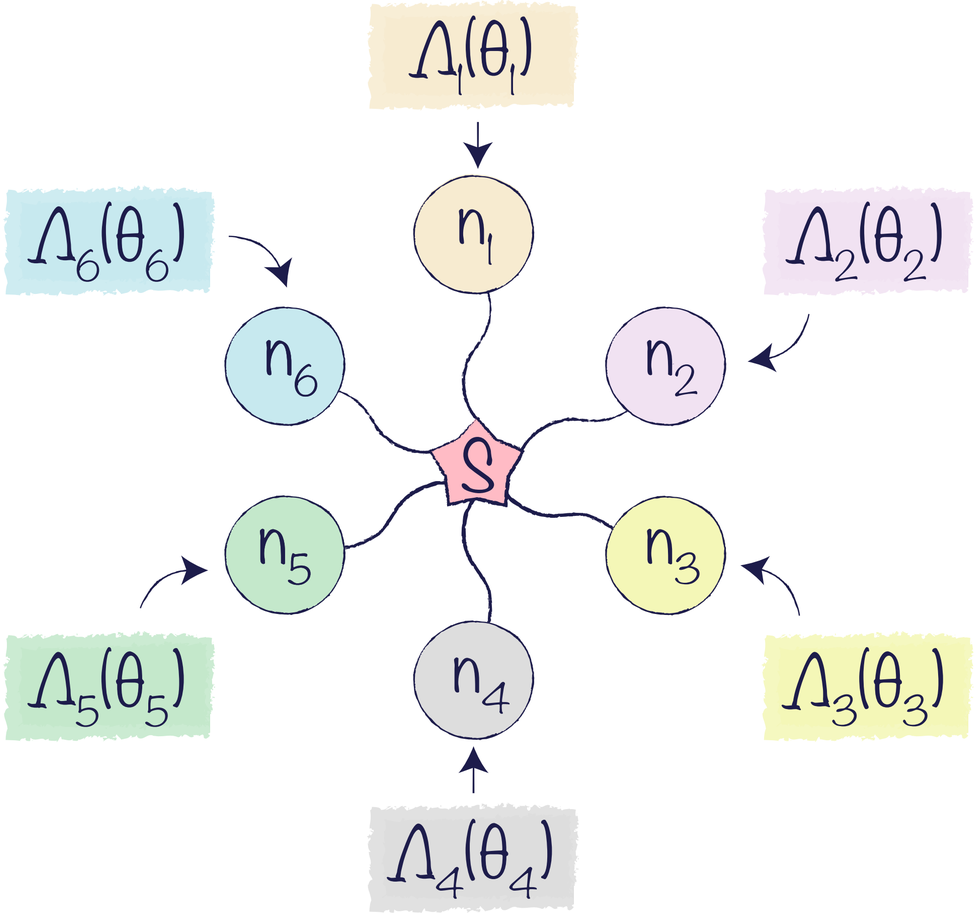
Example: average phase estimation

i. The source prepares:

$$|\mathrm{GHZ}_6\rangle = \frac{1}{\sqrt{2}}(|0\cdots0\rangle + |1\cdots1\rangle)$$

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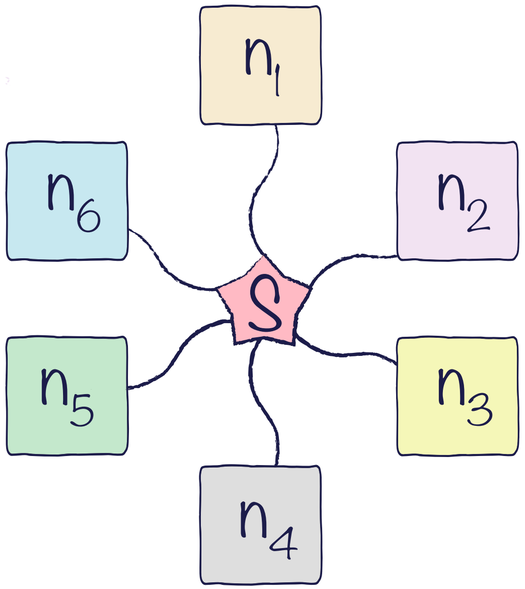
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$$U(\theta_j) = |0\rangle\langle 0| + e^{i\theta_j}\,|1\rangle\langle 1| \quad \text{with} \quad \mathbf{\Lambda_\theta} = \bigotimes_{j=1}^{6} U(\theta_j)$$

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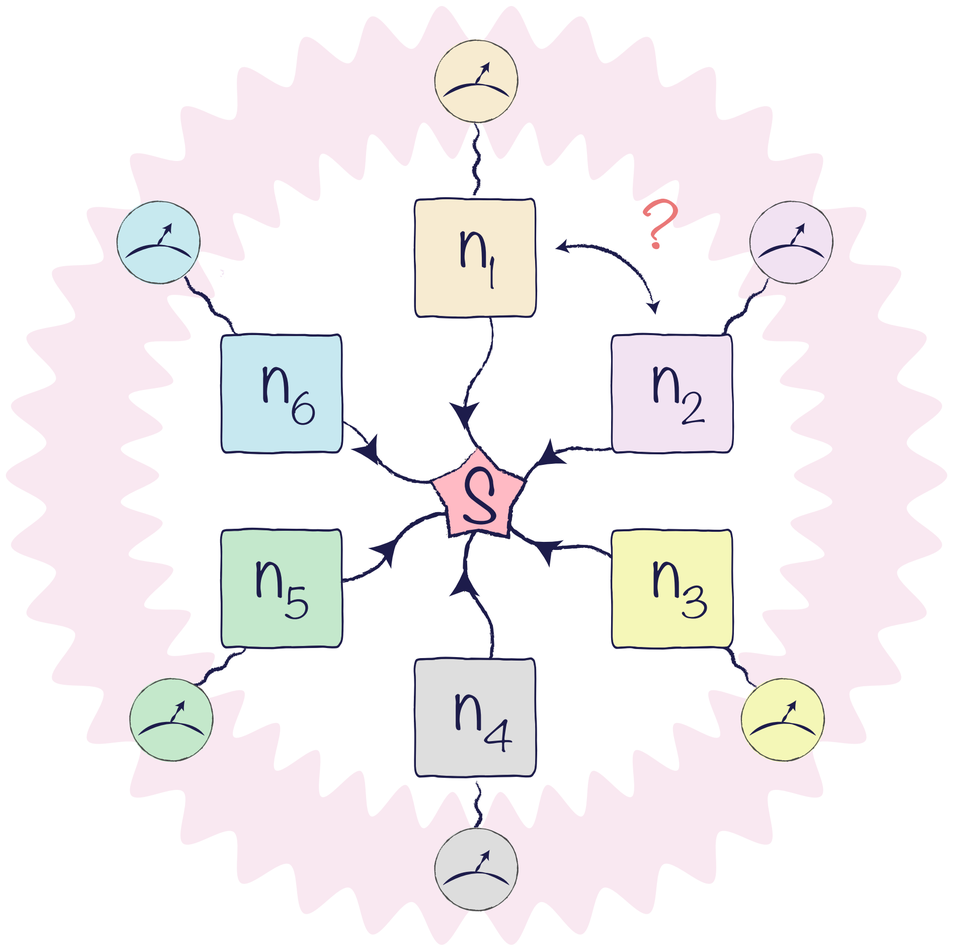
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iii. Each node measures $X \longrightarrow P(\pm|\theta)$

Do outcomes reveal ***only*** the *average phase?*

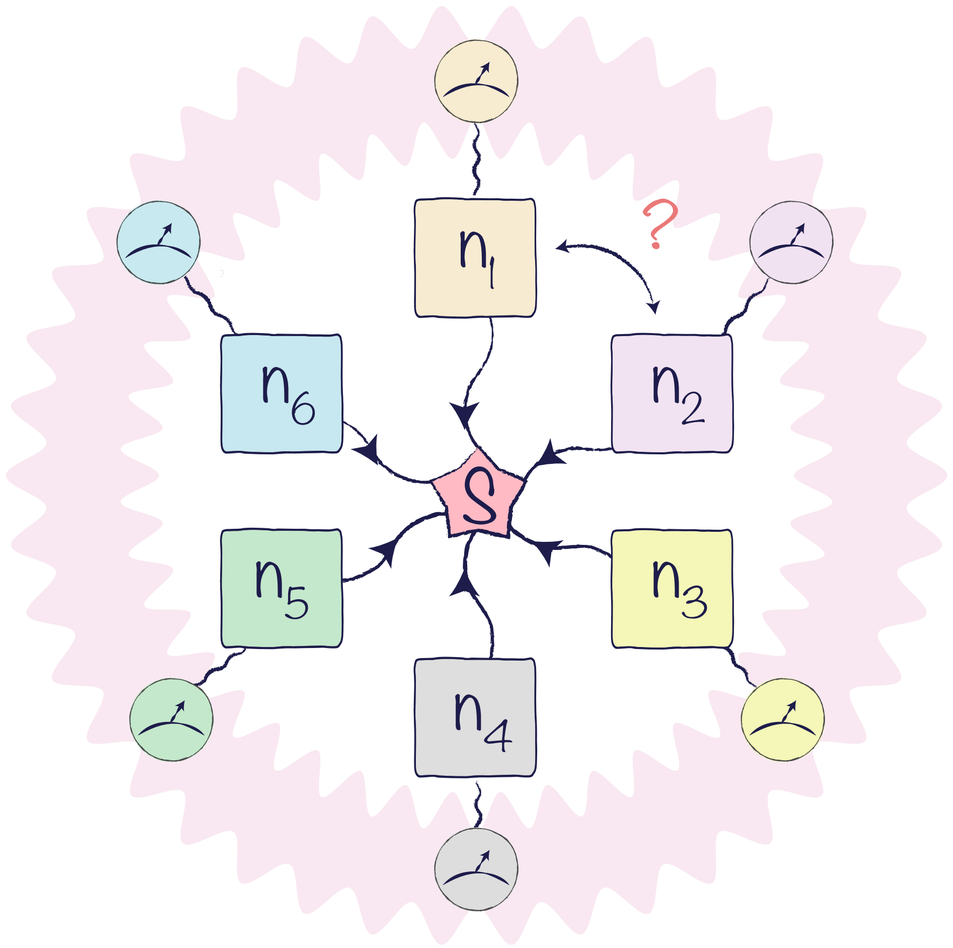
M Hassani, S Scheiner, MGA Paris, D Markham, Physical Review Letters **134**, 030802 (2025)

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- GHZ states are completely *private!*

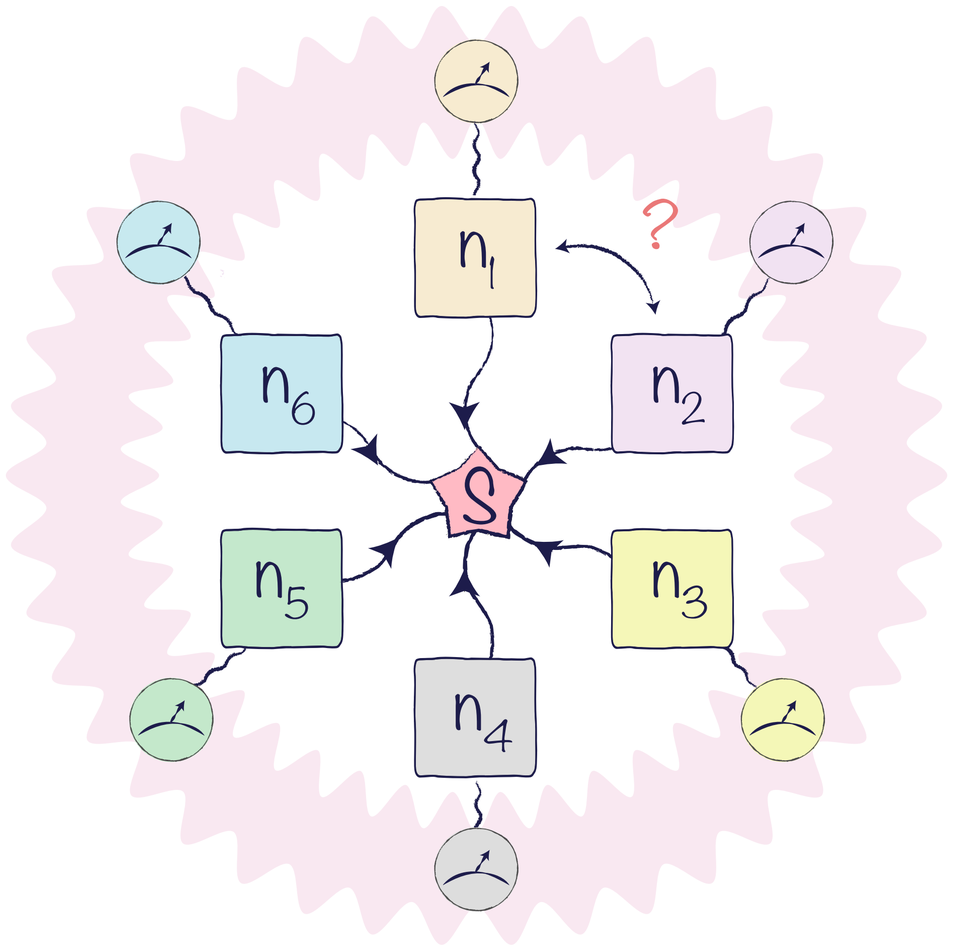
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i. The source prepares:

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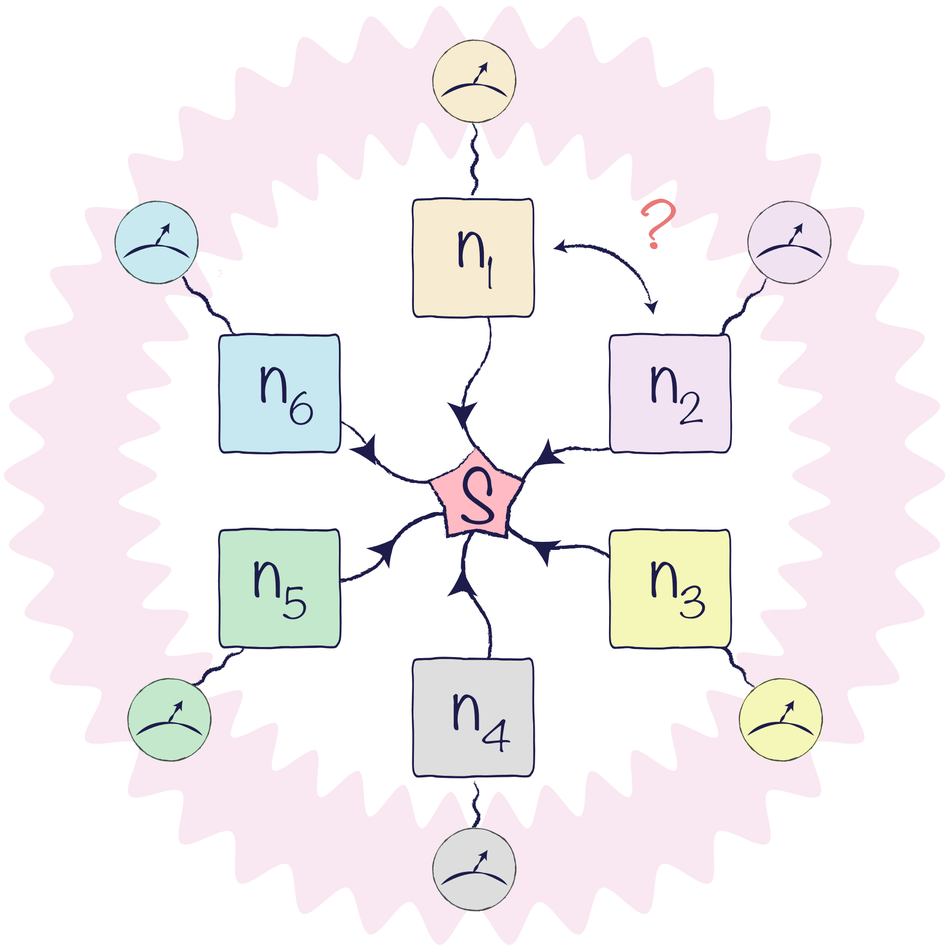
★ $|+\rangle^{\otimes 6}$ allows us to learn *individual* parameters!

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What about ***continuous-variable*** systems?

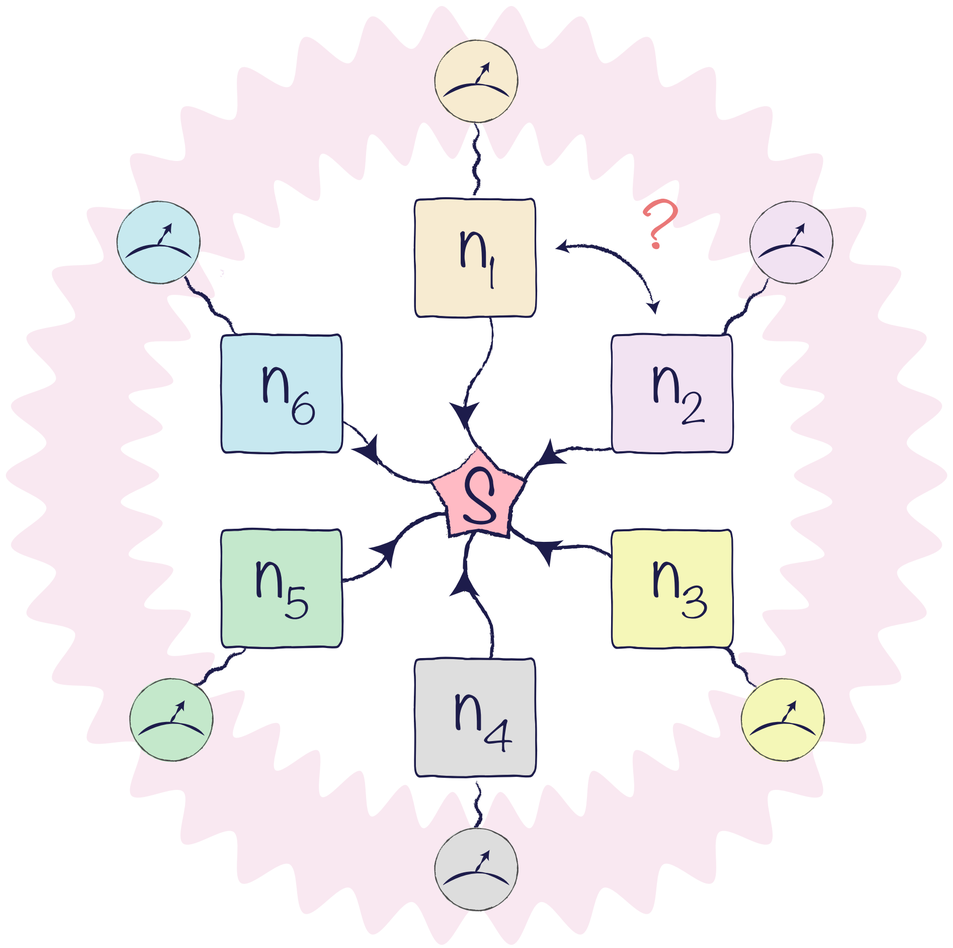
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... this lead us to ask:

- ... also allows us to learn *individual* parameters!

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Can **Gaussian** networks achieve **private** sensing with ***experimentally*** feasible resources?

II. When do sensors learn``*too much*''?

Privacy

- Multi-parameter estimation

Which *parameter directions* are *learnable* from $\hat{\rho}(\boldsymbol{\theta})$?

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anticommutator: $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$

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SLD: $2\partial_{\theta_\nu}\hat{\rho}(\boldsymbol{\theta}) = \hat{\rho}(\boldsymbol{\theta})\hat{L}_{\theta_\nu} + \hat{L}_{\theta_\nu}\hat{\rho}(\boldsymbol{\theta})$

- *parameters:* $\boldsymbol{\theta} = (\theta_1, \cdots, \theta_M)$
- *combination/directions:* $\boldsymbol{u}$ [e.g., $\boldsymbol{u} \propto (1, \cdots, 1)$]
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... and we *quantify* by: $P(\boldsymbol{Q}, \boldsymbol{v}_{\mathrm{avg}}) = \dfrac{\boldsymbol{v}_{\mathrm{avg}}^{\top}\boldsymbol{Q}\boldsymbol{v}_{\mathrm{avg}}}{\mathrm{tr}(\boldsymbol{Q})}$.

★ $0 \leq P(\boldsymbol{Q}, \boldsymbol{v}_{\mathrm{avg}}) \leq 1$

MGA Paris, Int. J. Quant. Inf. **7**, 125 (2009)

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III. Privacy in continuous-variable network

Gaussian network

- Bulding a network

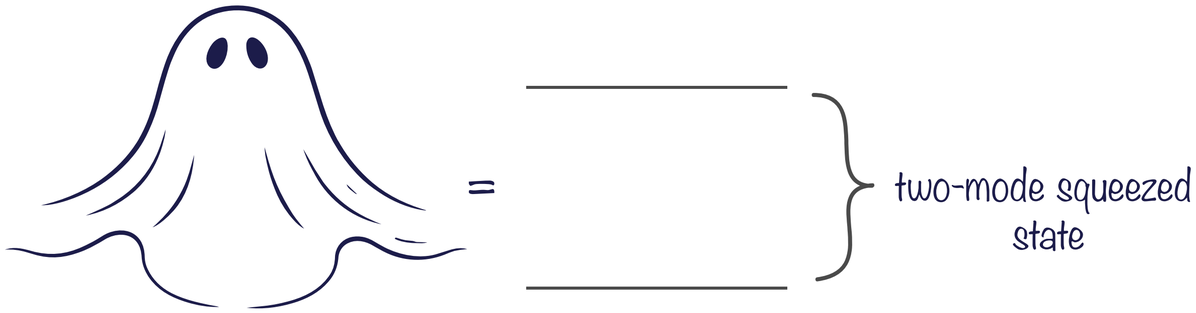
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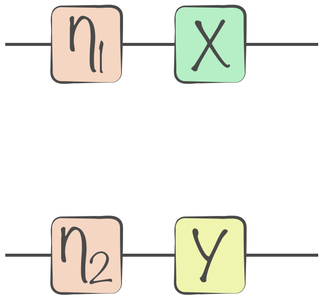
Gaussian network

- Bulding a network

What can't *Gaussian resources* + *linear optics* reveal about θ?



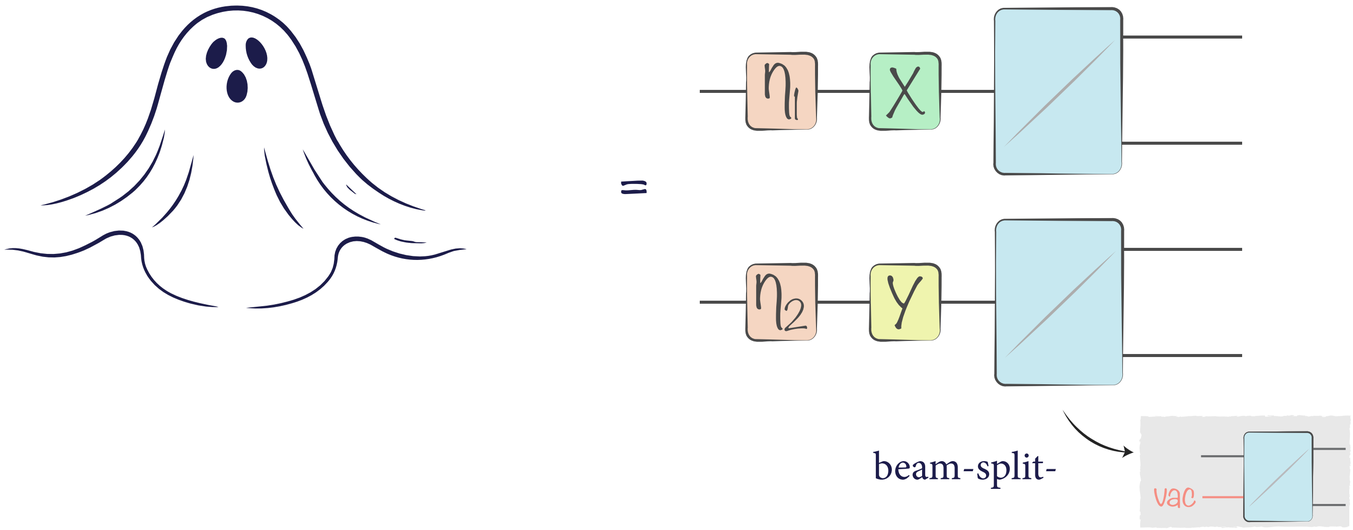
=



Gaussian network

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X Guo, C R Breum, J Borregaard, S Izumi, M V Larsen, T Gehring, M Christandl, J S Neergaard-Nielsen, U L Andersen
Nature Physics **16**, 281-284

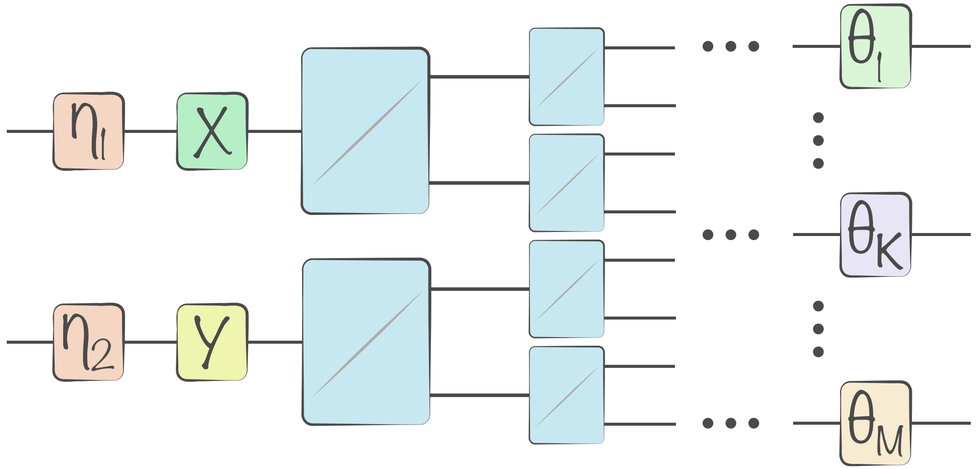
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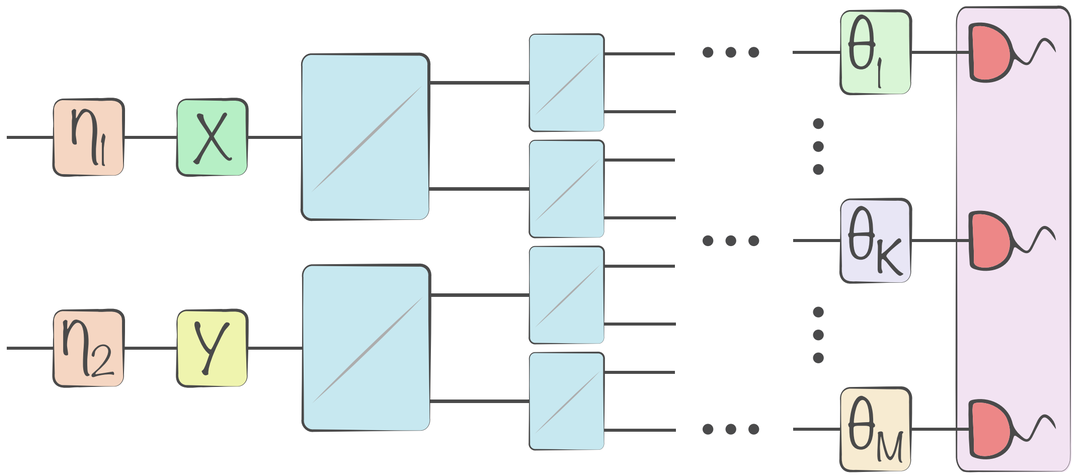
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fully analytical and experimentally

X Guo, C R Breum, J Borregaard, S Izumi, M V Larsen, T Gehring, M Christandl, J S Neergaard-Nielsen, U L Andersen
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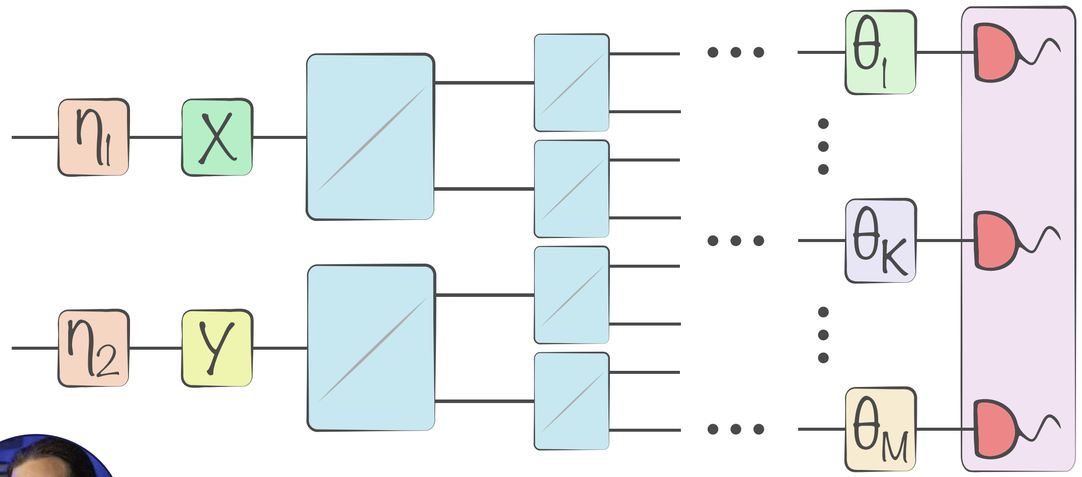
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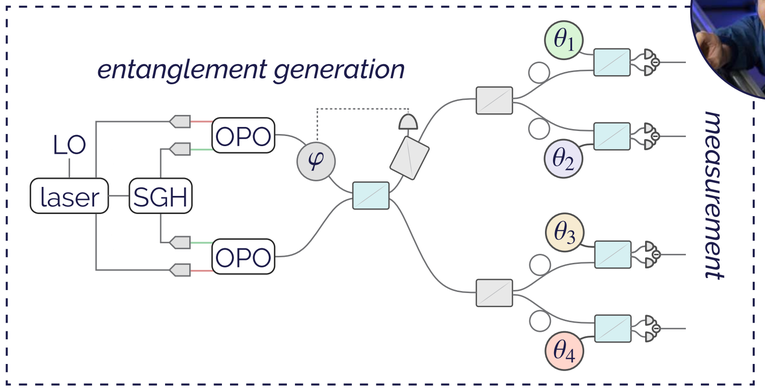
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fully analytical and experimentally

experiment in progress...





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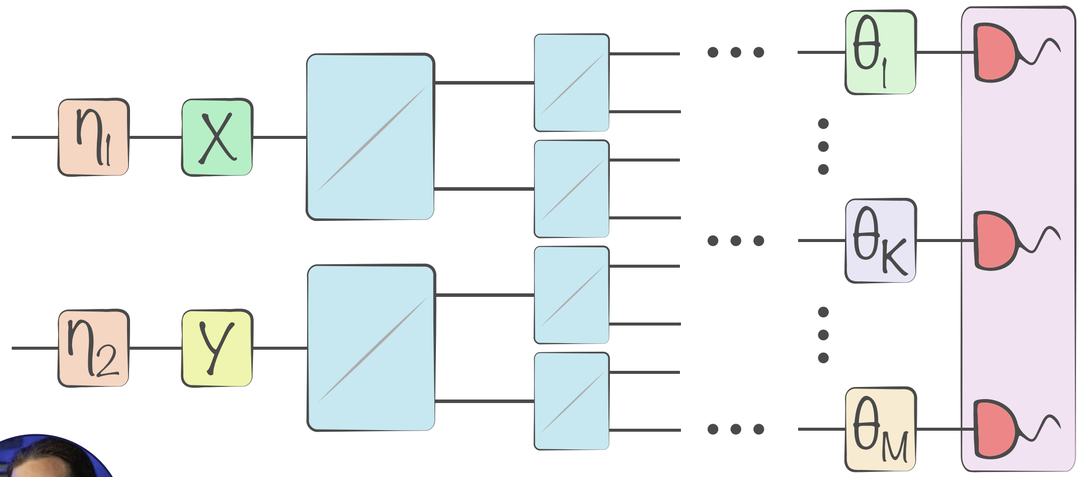
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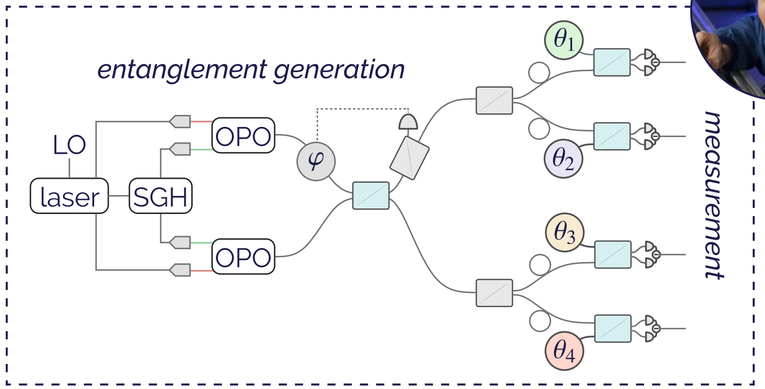
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fully analytical and experimentally

experiment in progress...





What's the ***privacy*** of this network?

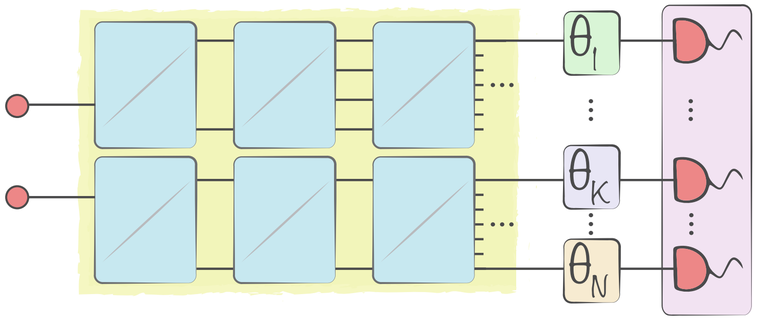
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Nature Physics **16**, 281-284

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Gaussian network

Multipartite network

What can we *learn* about this network?



⟶

★ *Analytical Q:* closed-form ⟶ *spectrum*

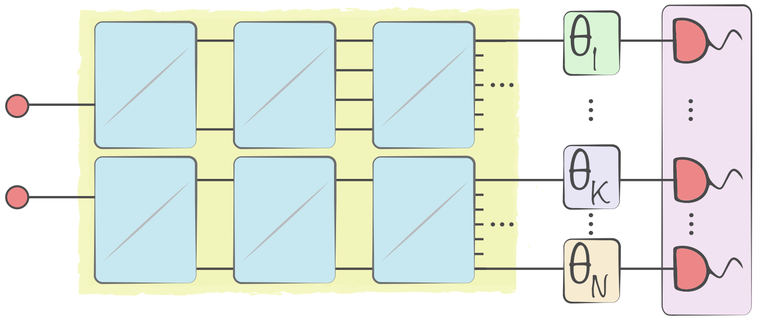
$$\lambda(\boldsymbol{Q}) = \left\{0,\ \underbrace{\frac{\cosh 2r - 1}{2^{N-2}}, \ldots, \frac{\cosh 2r - 1}{2^{N-2}}}_{M-2 \text{ times}},\ \frac{\cosh 4r - 1}{2^{N-1}}\right\}$$

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Gaussian network

Multipartite network

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★ *Analytical Q:* closed-form ⟶ *spectrum*

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"alternating"

"relative"

average

★ Unobservable direction: $s = (1, \cdots 1, -1 \cdots -1)$

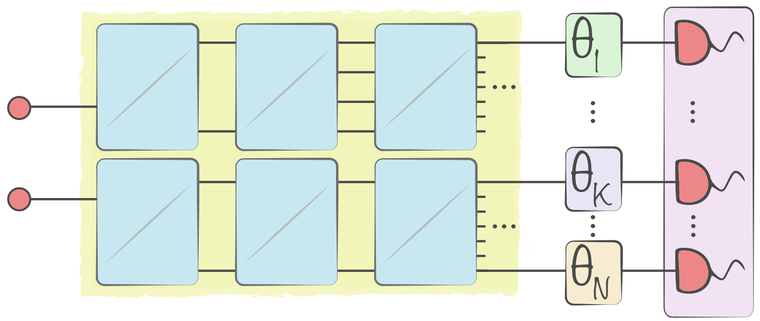
★ Each θ_j is *individually private!*

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Gaussian network

Multipartite network

What can we *learn* about this network?



- Unobservable direction: $s = (1, \cdots 1, -1 \cdots -1)$
- Each θ_j is *individually private!*

- *Analytical Q:* closed-form ⟶ *spectrum*

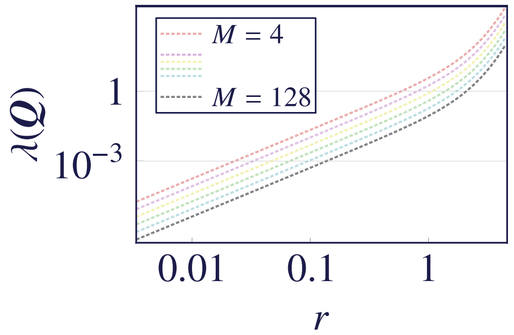
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"alternating"

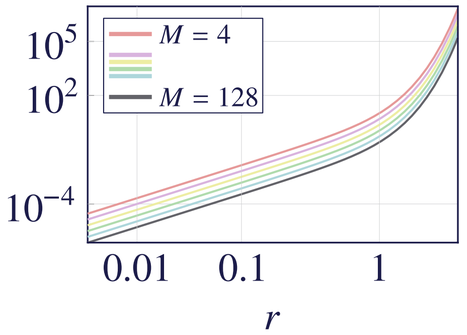
"relative"

average

shot-noise scaling



Heisenberg scalling

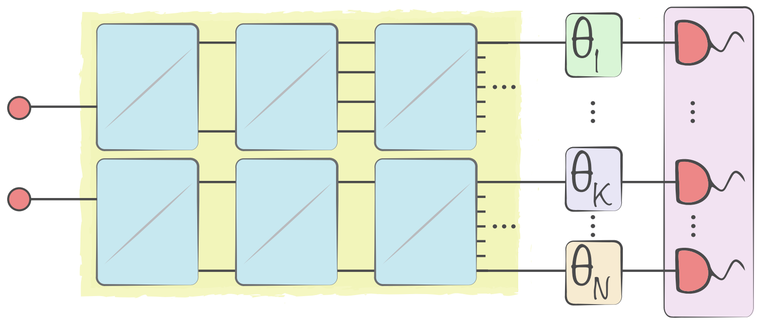


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Gaussian network

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★ Unobservable direction: $\boldsymbol{s} = (1, \cdots 1, -1 \cdots -1)$

★ Each θ_j is *individually private!*

What about the *privacy measure*?

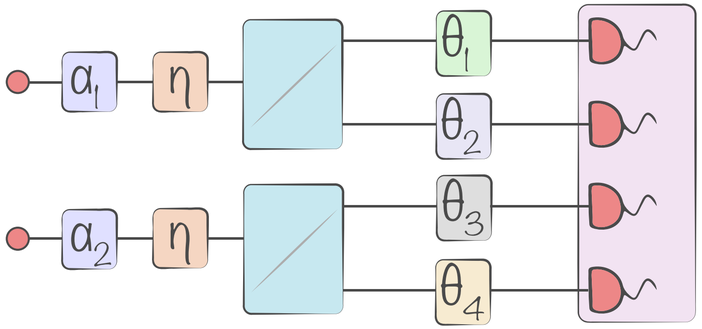
$$P(\boldsymbol{Q}, \boldsymbol{v}_{\text{avg}}) = 1 - \frac{M-2}{[(M-1) + \cosh 2r]}$$

Minimal scenario for privacy?

Gaussian network

- Minimal scenario

  - *Smallest* yet interesting network

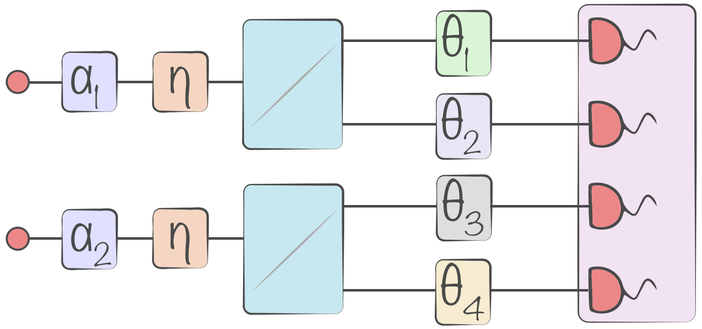


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Gaussian network

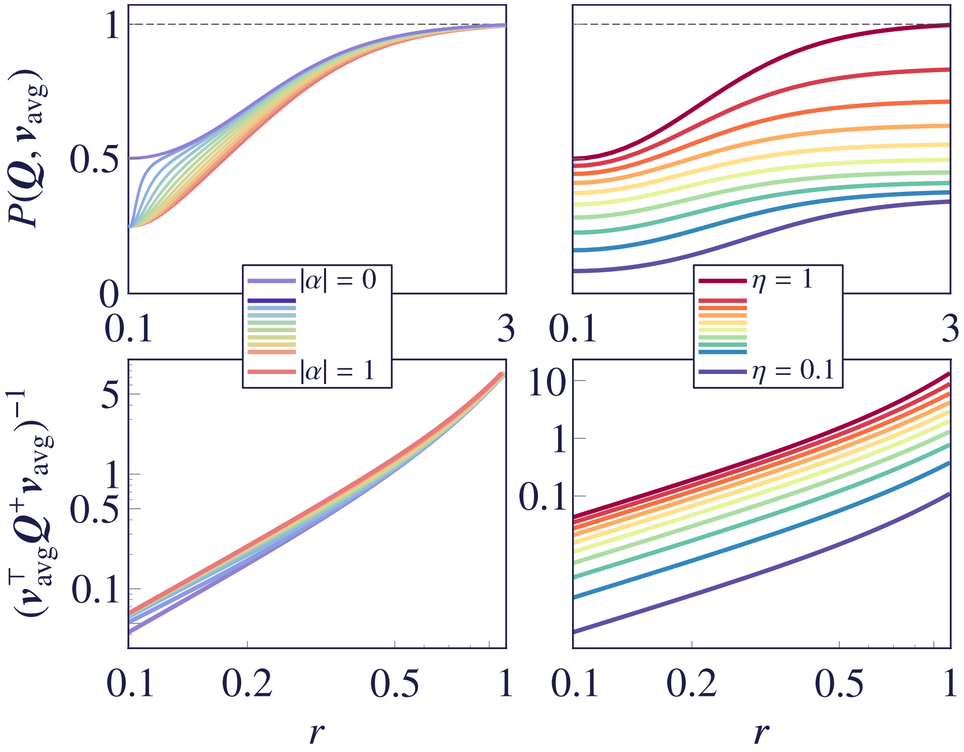
Minimal scenario

- *Smallest* yet interesting network



- *Strong* privacy only *asymptotically*...

$$P(r) = 1 - \frac{2}{3 + \cosh 2r}$$

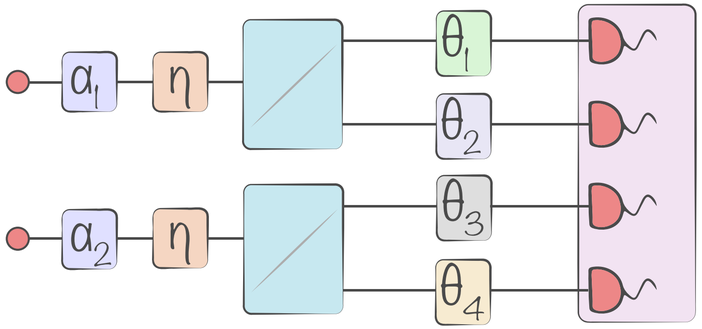


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Gaussian network

Minimal scenario

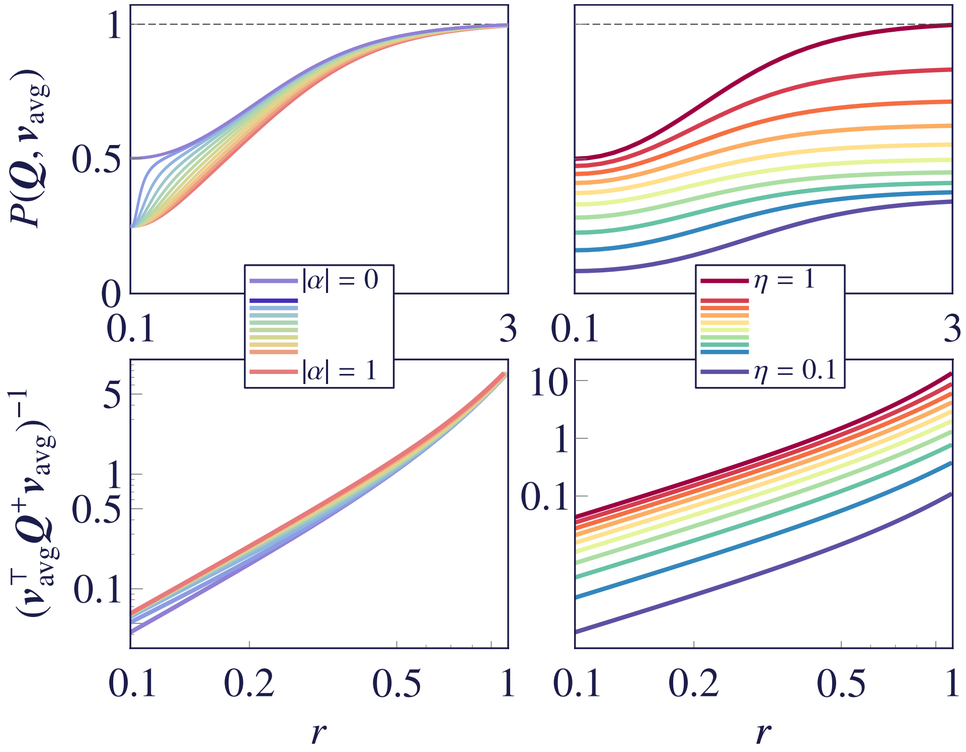
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- *Strong* privacy only *asymptotically*...

$$P(r) = 1 - \frac{2}{3 + \cosh 2r}$$

- *Optimal* parameters *under loss/displacement*

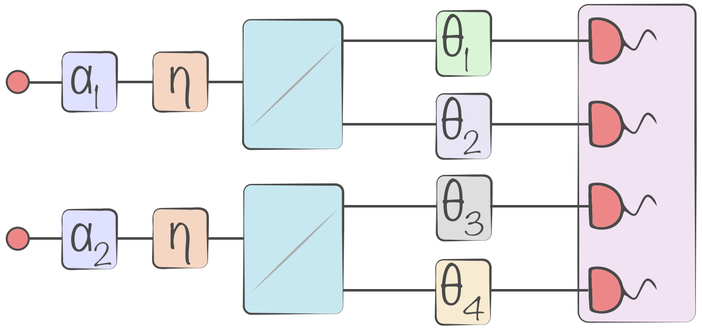


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Gaussian network

Minimal scenario

- *Smallest* yet interesting network

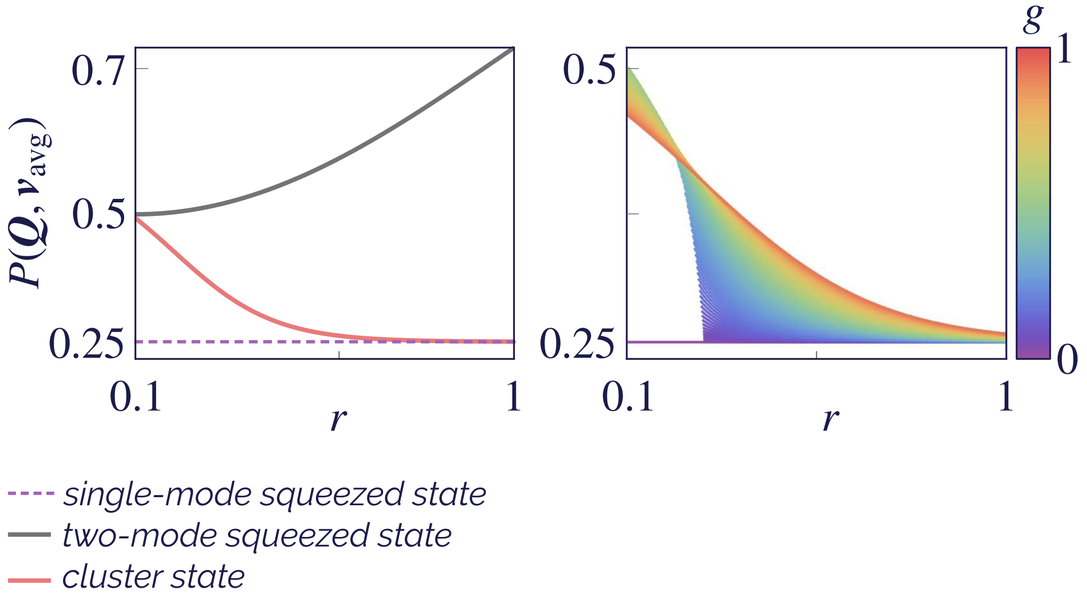


- *Strong* privacy only *asymptotically*...

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- *Optimal* parameters *under loss/displacement*

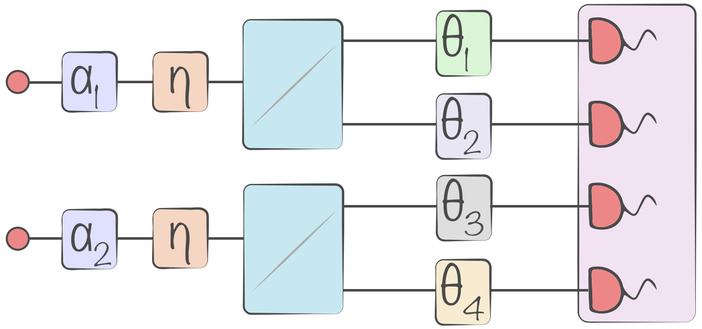
- *Protocol benchmark* with different *Gaussian states*



Gaussian network

Minimal scenario

- *Smallest* yet interesting network



- *Strong* privacy only *asymptotically*...

$$P(r) = 1 - \frac{2}{3 + \cosh 2r}$$

- *Optimal* parameters *under loss/displacement*

- *Protocol benchmark* with different *Gaussian states*

Privacy in Distributed Quantum Sensing with Gaussian Quantum Networks

Uesli Alushi[1,*] and Roberto Di Candia[1,†]

[1] *Department of Information and Communications Engineering, Aalto University, Espoo 02150, Finland*

We study the privacy properties of distributed quantum sensing protocols in a Gaussian quantum network, where each node encodes a parameter via a local phase shift. For networks with more than two nodes, achieving perfect privacy is possible only asymptotically, in the limit of large photon numbers. However, we show that optimized fully symmetric Gaussian states enable improved privacy levels while maintaining near-optimal sensing performance. We show that local homodyne detection achieves a quadratic scaling of precision with the total number of photons. We further analyze the impact of thermal noise in the preparation stage on both privacy and estimation precision. Our results pave the way for the development of practical, private distributed quantum sensing protocols in continuous-variable quantum networks.

U Alushi & R Di Candia, arXiv:**2509.22103**

Outlook

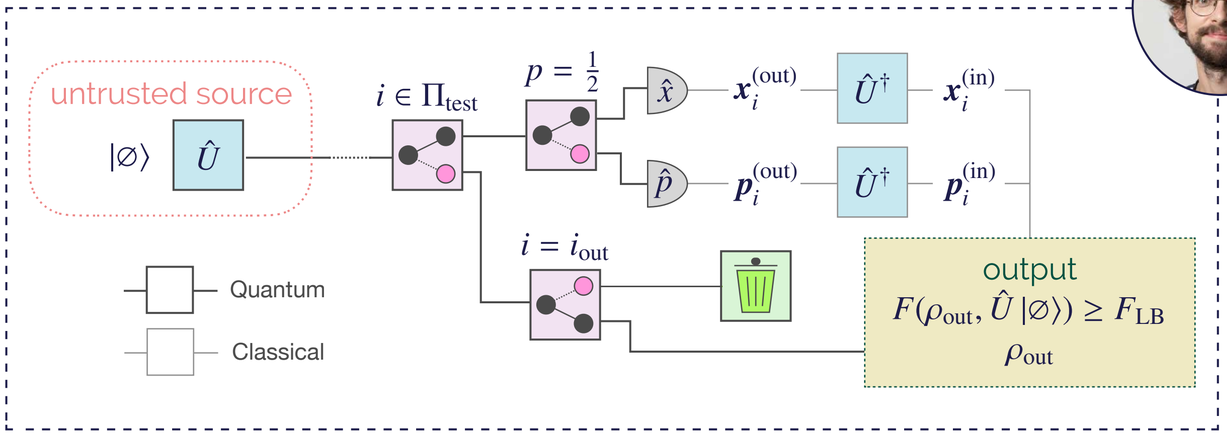
What I talked about:

- ***CV Protocol*** for privacy in CV-systems

What I haven't talked about:

- ***Certification protocol*** for privacy in CV-systems





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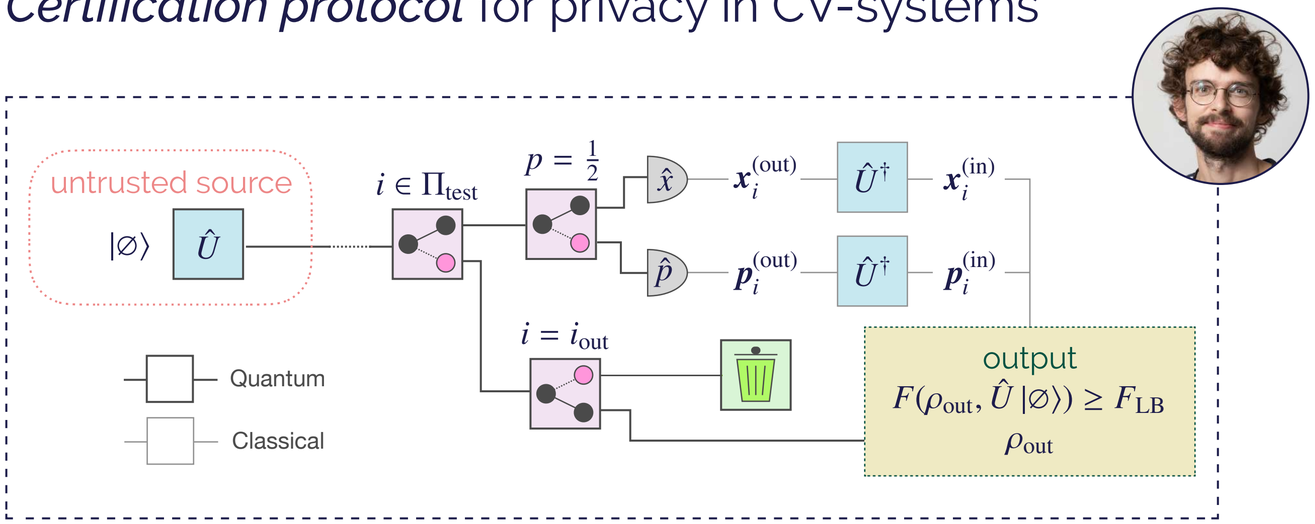
Outlook

What I talked about:

- ***CV Protocol*** for privacy in CV-systems

What I haven't talked about:

- ***Certification protocol*** for privacy in CV-systems



Where to head to?

Can we ***achieve strong privacy*** in CV-systems?

Do we have ***better*** Gaussian protocols?

